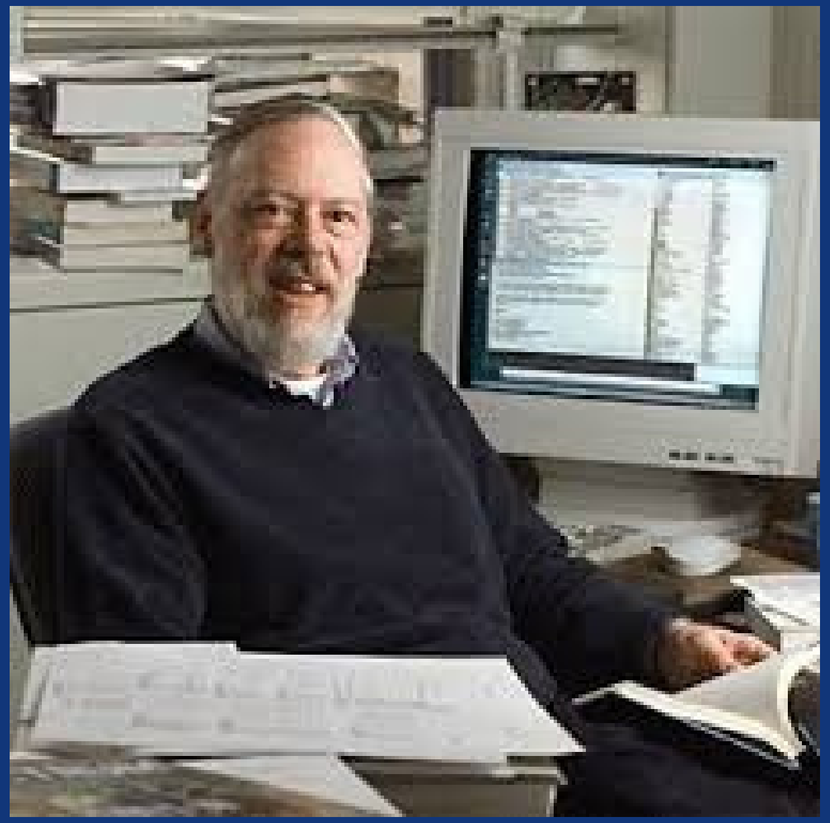
C
The Mother of All Languages

Praneel Chhabria-1262251874
Asmit Sen -1262251843
Aryaman Kumar -1262251780
Abdul Razzaque - 1262250672

The Foundation of Modern Computing

TheMotherof All Languages: Cis considered the"Mother" because it provided the fundamental syntax and structurefor nearly every modern language used today. A Shift in Programming: Before C, programmers mostly used Assembly (hard to read) or B (too limited). C found the "sweet spot" between human readability and machine speed.

Created by a Legend: Developed by Dennis Ritchie at Bell Labs in the early 1970s.



Building Operating System

C's critical role in OS development

Unix

The Unix operating system was **rewritten in C**, allowing for unprecedented portability across various hardware platforms, thereby revolutionizing software distribution and accessibility.

Windows Kernel

Key components of the Windows kernel are written in C, showcasing its **system-level capabilities** and making it integral to the performance and functionality of the operating system.

The Problem: Early operating systems were tied to specific hardware. If you changed the computer, you had to rewrite the OS.

The Solution: In 1973, the Unix kernel was rewritten in C.

Influence of C on Modern Programming Languages

C++

Directly extends C by adding object orientation.

Java

Inspired by C's syntax and structure for readability.

Python

Borrowed C-like syntax elements for better clarity.

Performance and Relevance

Why C remains vital today

- Close-to-Metal Programming: C allows developers to work directly with memory addresses and hardware.
- Resource Management: In environments with limited RAM or battery life—like IoT devices, car ECUs, or medical equipment—C's efficiency is unmatched.
- Performance: For high-speed applications like game engines, database systems, and AI compilers, C remains the gold standard for speed.

Key Features of the C Programming Language

- ***Procedural and structured language*** – Programs are written using functions and logical blocks, making large programs easier to manage.
- ***Rich set of data types and operators*** – Supports basic types (int, char, float, double) and powerful operators for low-level manipulation.
- ***Pointers and direct memory access*** – Allows efficient handling of memory, arrays and data structures.
- ***Portable language*** – C programs can be compiled on many different platforms with minimal changes.
- ***Extensible using libraries*** – Developers can create and reuse their own libraries.

Why C is Ideal for System Programming

→ Direct interaction with hardware using memory addresses and registers.

→ Minimal runtime overhead, making programs fast and lightweight.

→ Fine control over memory allocation and deallocation using malloc() and free().

→ Widely used for building kernels, drivers and embedded software.

→ Acts as a bridge between high-level languages and assembly language, combining readability with performance.

Role of C in Embedded Systems and Electronics

→ C is the primary language for microcontrollers such as Arduino, PIC and ARM-based controllers.

→ Used in automotive ECUs, medical devices, sensors and robotics systems.

→ Small memory footprint makes C suitable for devices with limited RAM and storage.

→ Real-time applications benefit from predictable execution and low latency.

Learning C: Foundation for Future Languages

Understanding C helps in learning **C++, Java**, **Python** and many other modern languages more easily.

→ ***Concepts such as***:
→ Variables and data types
→ Loops and conditionals
→ Functions and modular programming
→ Memory and pointers
→ Form the core foundation of programming.

→ ***C improves a programmer's ability to***:
→ Write efficient code
→ Debug low-level errors
→ Understand how programs run inside a computer.

→ ***Many computer science subjects such as***:
→ operating systems
→ compilers
→ computer architecture
rely heavily on C concepts.

Thank

You